

Adelfa Class Record

I. Problem Statement

Teachers are busy people, even behind the scenes. They have to prepare lesson plans, presentation slides, and a comfortable classroom for their students. They also have to manage and organize a lot of student information, including class records. Grade-8 Adelfa's adviser wants to keep track of the section's class records, being able to access each student's information and change it. As technology continues to become more globalized, the need for faster and more organized ways for teachers to manage student data also continues to grow.

II. Problem Identification

The traditional paperwork is too unreliable and time-consuming for accurate data. There are already existing spreadsheets that can accommodate these requirements, but they can also be inefficient and inaccurate when done manually.

III. Problem Decomposition

The following are the specific problems teachers face with traditional or spreadsheet student records:

- **Manual Data Entry**
Teachers type or write all data one by one, making it tiring, time-consuming, and possibly inconsistent.
- **Calculation Errors**
With records like report cards, teachers might make some mistakes with computations.
- **Poor Data Organization**
Files may be stored in random drawers, folders, or notebooks due to limited space, and they may potentially be lost, and searching for them may be a hassle.
- **Difficulty in Accessing Records**
Especially with the traditional way of managing files, finding specific records can be slow and inconvenient. This is also a big problem for teachers who handle a large class or a lot of students in general.

IV. Proposed Solution

To address the identified problems, our group decided to propose a program for Adelfa's class adviser, which is the "Adelfa Class Records." This program's main purpose is to aid the teacher in searching for, accessing, managing, and computing Adelfa's student records.

The following are the specific requirements or features of the program:

- The program should allow the teacher to search for a student record using a specific key, such as the student's class number. This will make locating a student's data easier.
- The program should be organized not only in its formatting, but also in its storage of information. They should be easily accessible by grouping them accordingly, like the performance of each student per subject.
- The program should allow the teacher to edit information easily. For instance, it lets the teacher edit a student's scholarship status, add a remark about a student's performance in class, and remove other stuff.
- The program should include a calculator, allowing the teacher to just input the necessary data for the computer to calculate a student's grade and complete their report card. Summaries or a detailed report can be displayed, depending on the teacher's needs.

V. Feasibility and Scope

The feasibility of this program is not that strict—it can be developed in simple programming languages such as Python or Java. Moreover, not only can it be used by the teacher, but also by the students, with more limitations on what they can access.

However, this is only fit for a classroom-wide management, specifically for the Adelfa section and its teacher. The programmers will focus on developing the key requirements of the system:

- Student search through a key
- Organized database of student information
- Easy management of student information and database sections
- Grades calculator
- Report card and summary of student information generation

VI. Project Overview

Adelfa Class Records is a program designed to help the adviser of grade-eight Adelfa keep track of student records smoothly. The program aims to simplify the work of the user (the teacher) to avoid messy, incomplete, or disorganized records. It keeps and remembers management information, such as attendance, the number of absences, the number of times the student has an incomplete uniform, etc.

VII. Methodology

The programming language used in this project is Python to simplify the coding process. The project includes input variables, match-case, and a while loop. The program works first by gathering data and storing it in the database. Then, it matches the input of the user to the stored variables to find the data gathered. Lastly, the output shows the input, and the program saves the new data.

VIII. Objectives

The program aims to aid Grade-8 Adelfa's adviser in keeping track of and managing student records by ensuring the following are met and made easier to accomplish:

- The teacher can access student information easily.
- The teacher can calculate grades without too much worry of making mistakes and taking too long.
- Student records are organized and won't be easily lost.
- The teacher will become less stressed.